



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY  
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

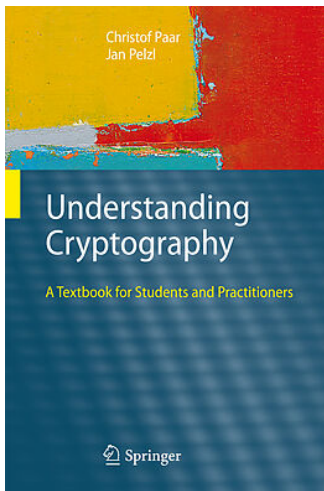
# Introduction to Cryptography and Security

## The Data Encryption Standard (DES) and Alternatives

September 27, 2024

# Textbook

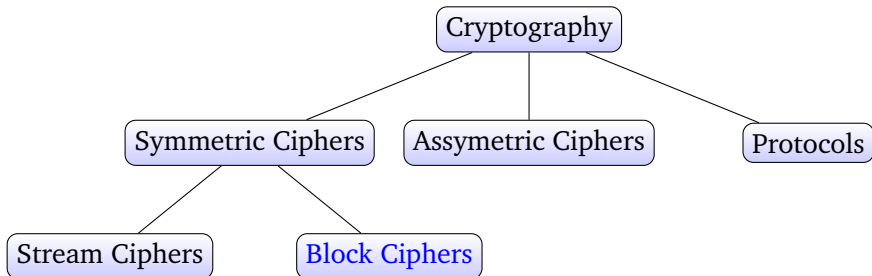
<https://www.crypto-textbook.com>



# Outline

- 1 Introduction
- 2 Overview of the DES Algorithm
- 3 Internal Structure of DES
- 4 Key Schedule
- 5 Decryption
- 6 Security of DES

# Cryptography



# Introduction to DES

- Proposed by IBM in 1974 based on the cipher *Lucifer*.

*Lucifer is a Feistel cipher which encrypts blocks of 64 bits using a key size of 128 bits.*

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- It seems certain that the National Security Agency (NSA) influenced changes to the cipher, which was rechristened DES.

# Introduction to DES

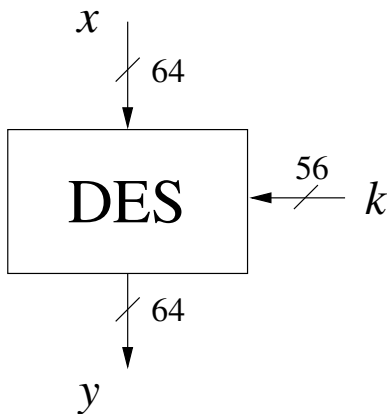
- Proposed by IBM in 1974 based on the cipher *Lucifer*.

*Lucifer is a Feistel cipher which encrypts blocks of 64 bits using a key size of 128 bits.*

- It seems certain that the National Security Agency (NSA) influenced changes to the cipher, which was rechristened DES.
- One of the changes that occurred was that DES is specifically designed to withstand differential cryptanalysis, an attack not known to the public until 1990.

## Introduction to DES 2

- Allegedly, the NSA also convinced IBM to **reduce the Lucifer key length of 128 bit to 56 bit**, which made the cipher much more vulnerable to brute-force attacks.
- Some people conjectured that the NSA would be able to search through a key space of  $2^{56}$ , thus breaking it by brute-force.
- Despite of all the criticism and concerns, in 1977 the National Bureau of Standard (NBS) finally released all specifications of the modified IBM cipher as the Data Encryption Standard to the public.



- Currently, DES is no longer secure due to the short key size.
- But 3DES is secure.

# Principle of building block cipher

According to Claude Shannon

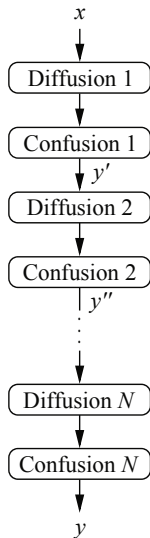
- **Confusion** is an encryption operation where the relationship between key and ciphertext is obscured.
- **Diffusion** is an encryption operation where the influence of one plaintext symbol is spread over many ciphertext symbols with the goal of hiding statistical properties of the plaintext.

## Example

Principle of diffusion of a block cipher



## Principle of building block cipher

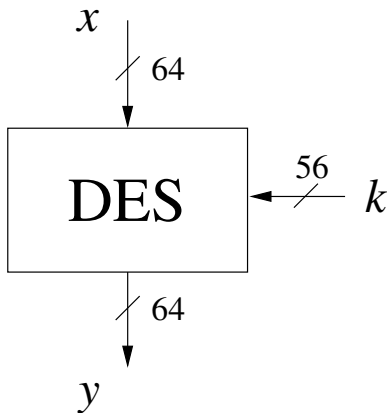


Principle of an  $N$  round product cipher, where each round performs a confusion and diffusion operation

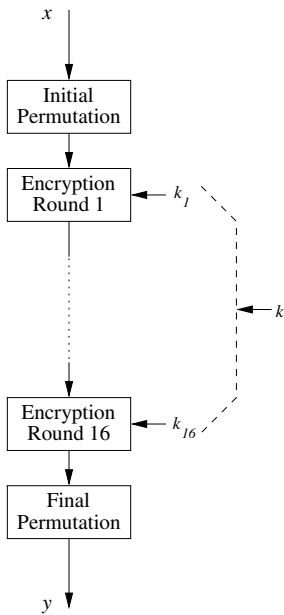
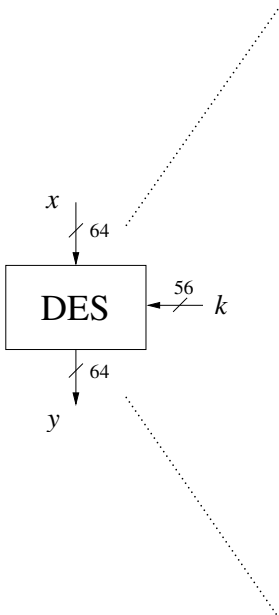
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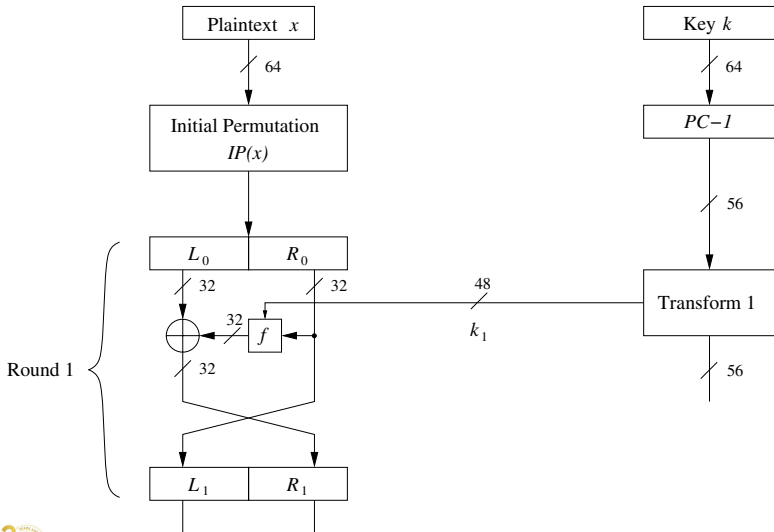
# DES



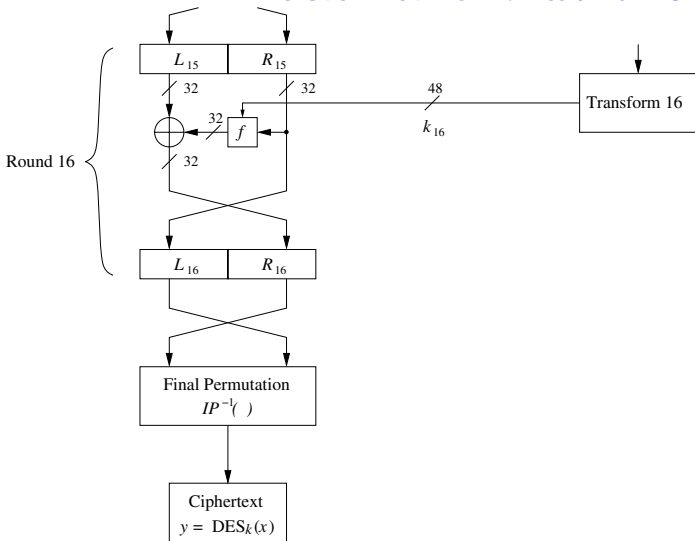
- The block size is 64 bits
- The key length is 56 bits.



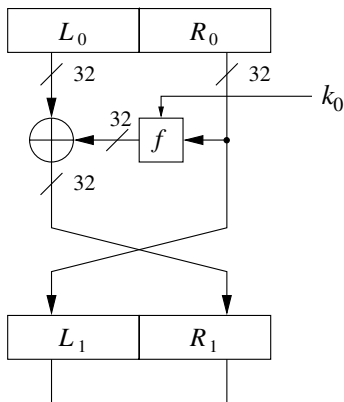
# Feistel network: Round 1



## Feistel network: Round 16



# The Feistel structure of DES



- In symbols:

$$L_i = R_{i-1}$$

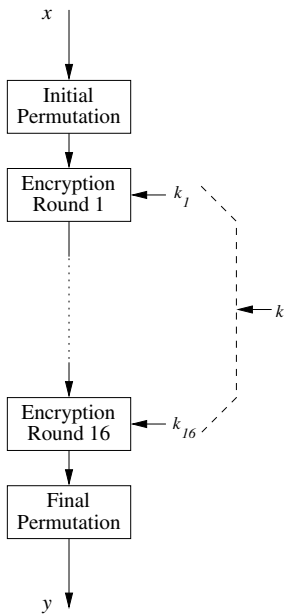
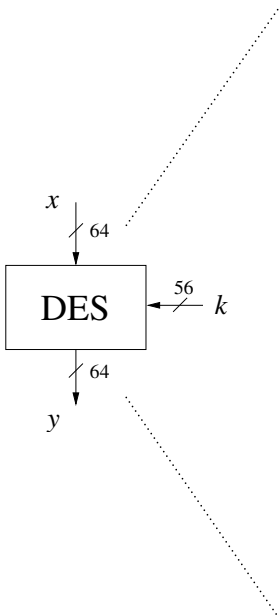
$$R_i = L_{i-1} \oplus f(R_{i-1}, k_i)$$

- How to compute the inverses?

$$(L_i, R_i) \longrightarrow (L_{i-1}, R_{i-1})$$

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## Initial and Final Permutation

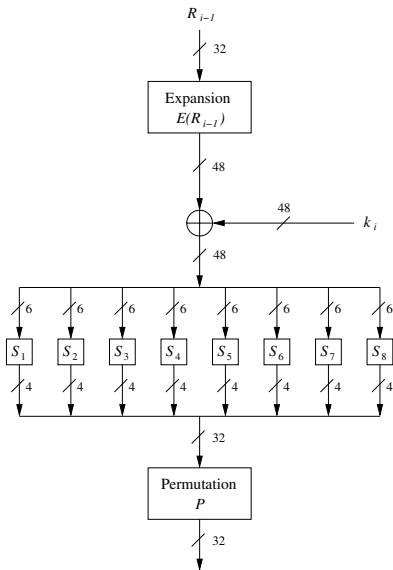
$IP$

|    |    |    |    |    |    |    |   |
|----|----|----|----|----|----|----|---|
| 58 | 50 | 42 | 34 | 26 | 18 | 10 | 2 |
| 60 | 52 | 44 | 36 | 28 | 20 | 12 | 4 |
| 62 | 54 | 46 | 38 | 30 | 22 | 14 | 6 |
| 64 | 56 | 48 | 40 | 32 | 24 | 16 | 8 |
| 57 | 49 | 41 | 33 | 25 | 17 | 9  | 1 |
| 59 | 51 | 43 | 35 | 27 | 19 | 11 | 3 |
| 61 | 53 | 45 | 37 | 29 | 21 | 13 | 5 |
| 63 | 55 | 47 | 39 | 31 | 23 | 15 | 7 |

$IP^{-1}$

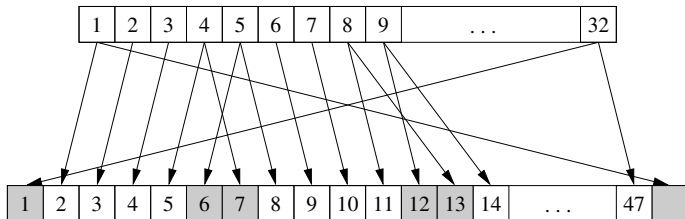
|    |   |    |    |    |    |    |    |
|----|---|----|----|----|----|----|----|
| 40 | 8 | 48 | 16 | 56 | 24 | 64 | 32 |
| 39 | 7 | 47 | 15 | 55 | 23 | 63 | 31 |
| 38 | 6 | 46 | 14 | 54 | 22 | 62 | 30 |
| 37 | 5 | 45 | 13 | 53 | 21 | 61 | 29 |
| 36 | 4 | 44 | 12 | 52 | 20 | 60 | 28 |
| 35 | 3 | 43 | 11 | 51 | 19 | 59 | 27 |
| 34 | 2 | 42 | 10 | 50 | 18 | 58 | 26 |
| 33 | 1 | 41 | 9  | 49 | 17 | 57 | 25 |

## The $f$ – function



- Expansion  $E(R_{i+1})$
- XOR with current round key  $i$
- substitution boxes  $S$ -box
- $P$  permutation

## Expansion $E$



| $E$ |    |    |    |    |    |  |
|-----|----|----|----|----|----|--|
| 32  | 1  | 2  | 3  | 4  | 5  |  |
| 4   | 5  | 6  | 7  | 8  | 9  |  |
| 8   | 9  | 10 | 11 | 12 | 13 |  |
| 12  | 13 | 14 | 15 | 16 | 17 |  |
| 16  | 17 | 18 | 19 | 20 | 21 |  |
| 20  | 21 | 22 | 23 | 24 | 25 |  |
| 24  | 25 | 26 | 27 | 28 | 29 |  |
| 28  | 29 | 30 | 31 | 32 | 1  |  |

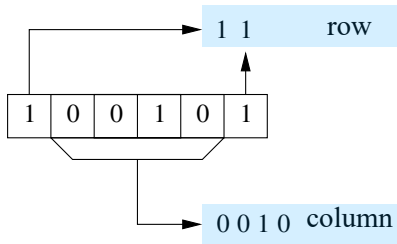
## S-Box

- S-box is a lookup table that maps a 6-bit input to a 4-bit output.
- DES has 8 S-boxes that are nonlinear functions

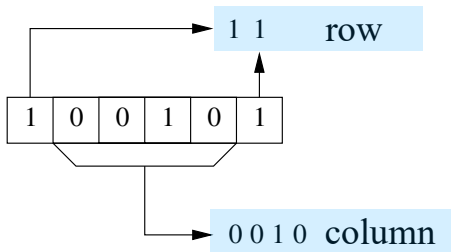
$$S(a) \oplus S(b) \neq S(a \oplus b).$$

(against differential cryptanalysis)

- The S-box is decrypted in a special way:



## S-box



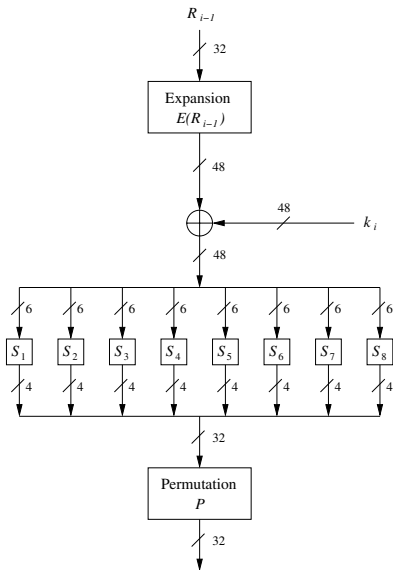
| $S_1$ | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0     | 14 | 04 | 13 | 01 | 02 | 15 | 11 | 08 | 03 | 10 | 06 | 12 | 05 | 09 | 00 | 07 |
| 1     | 00 | 15 | 07 | 04 | 14 | 02 | 13 | 01 | 10 | 06 | 12 | 11 | 09 | 05 | 03 | 08 |
| 2     | 04 | 01 | 14 | 08 | 13 | 06 | 02 | 11 | 15 | 12 | 09 | 07 | 03 | 10 | 05 | 00 |
| 3     | 15 | 12 | 08 | 02 | 04 | 09 | 01 | 07 | 05 | 11 | 03 | 14 | 10 | 00 | 06 | 13 |

## $P$ Permutation

| $P$ |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|
| 16  | 7  | 20 | 21 | 29 | 12 | 28 | 17 |
| 1   | 15 | 23 | 26 | 5  | 18 | 31 | 10 |
| 2   | 8  | 24 | 14 | 32 | 27 | 3  | 9  |
| 19  | 13 | 30 | 6  | 22 | 11 | 4  | 25 |

**Figure:** The permutation  $P$  introduces diffusion because the four output bits of each S-box are permuted in such a way that they affect several different S-boxes in the following round.

## The $f$ – function

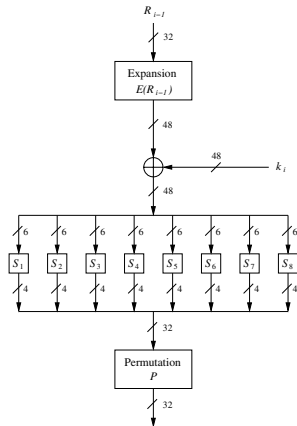
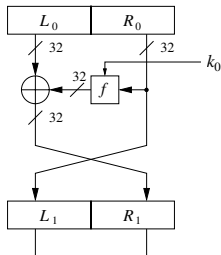
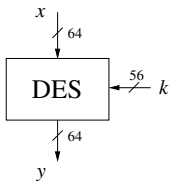


- Expansion  $E(R_{i+1})$
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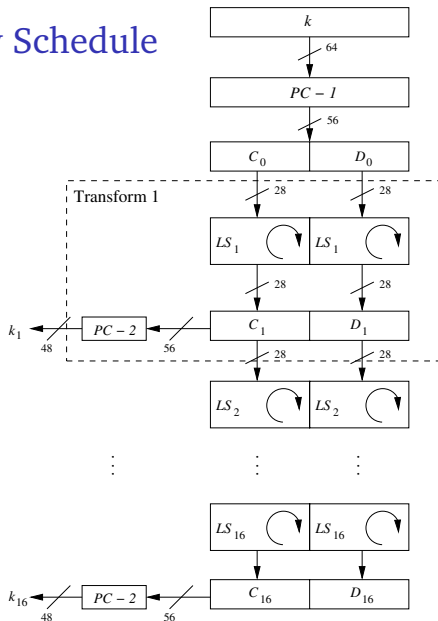
## Recall: Internal Structure of DES



# Key Schedule

- **Question:** How to calculate 16 subkey  $k_1, \dots, k_{16}$ ?
- Key schedule uses only simple operations (permutations and left rotation) on bits.

# Key Schedule

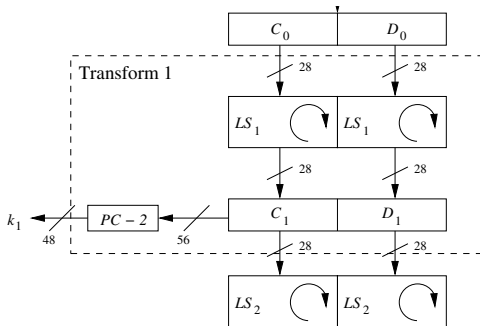


## Initial Key Permutation PC-1

- Remove the 8, 16, 24, ..., 64 bits of key  $k$  of size 64 bits.
- The real DES key is just  $(64 - 8) = 56$  bits.

| $PC - 1$ |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|
| 57       | 49 | 41 | 33 | 25 | 17 | 9  | 1  |
| 58       | 50 | 42 | 34 | 26 | 18 | 10 | 2  |
| 59       | 51 | 43 | 35 | 27 | 19 | 11 | 3  |
| 60       | 52 | 44 | 36 | 63 | 55 | 47 | 39 |
| 31       | 23 | 15 | 7  | 62 | 54 | 46 | 38 |
| 30       | 22 | 14 | 6  | 61 | 53 | 45 | 37 |
| 29       | 21 | 13 | 5  | 28 | 20 | 12 | 4  |

## $LS_i$ : Left shift (left rotate)



$$LS_i = \begin{cases} \text{Left Rotate 1 position} & \text{if } i = 1, 2, 9, 16 \\ \text{Left rotate 2 positions} & \text{otherwise.} \end{cases}$$

**Remark:** Total number of bits rotated is  $4 \times 1 + 12 \times 2 = 28$ , thus

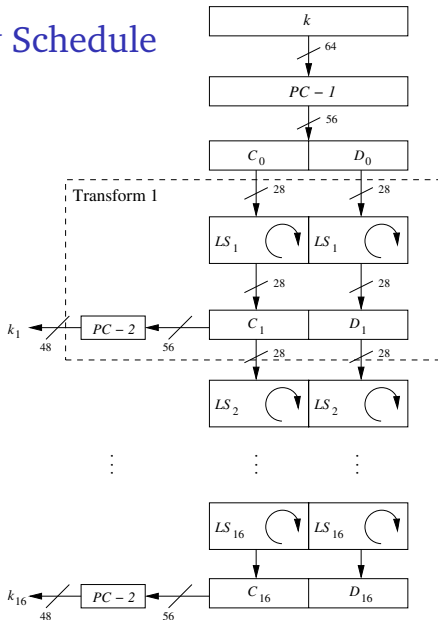
$$C_{16} = C_0; D_{16} = D_0.$$

## Round Key Permutation PC-2

- Remove 8 bits of  $\boxed{C_i} \boxed{D_i}$  ;
- The bit length of subkey  $k_i$  is  $56 - 8 = 48$  bit

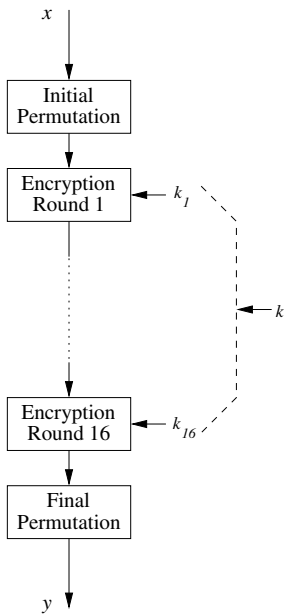
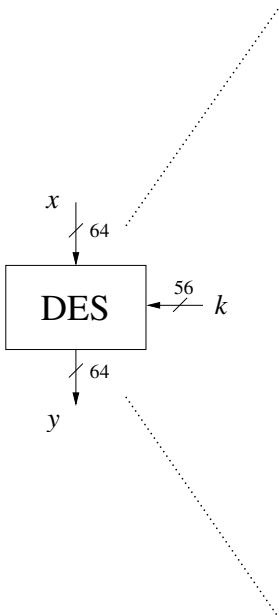
| <i>PC - 2</i> |    |    |    |    |    |    |    |
|---------------|----|----|----|----|----|----|----|
| 14            | 17 | 11 | 24 | 1  | 5  | 3  | 28 |
| 15            | 6  | 21 | 10 | 23 | 19 | 12 | 4  |
| 26            | 8  | 16 | 7  | 27 | 20 | 13 | 2  |
| 41            | 52 | 31 | 37 | 47 | 55 | 30 | 40 |
| 51            | 45 | 33 | 48 | 44 | 49 | 39 | 56 |
| 34            | 53 | 46 | 42 | 50 | 36 | 29 | 32 |

## Key Schedule

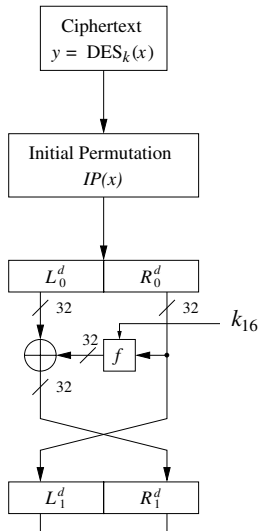
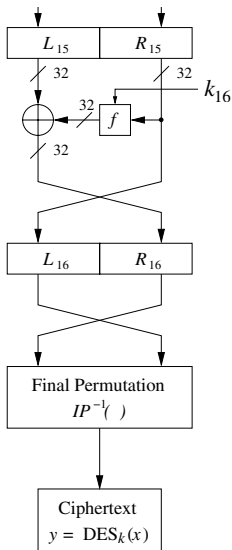


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# DES decryption



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# Exhaustive Key Search

## Problem

- given a few input output pairs

$$(x_i, y_i = \text{Enc}(k, x_i))$$

for  $i = 1, 2, 3$ .

- find key  $k$ .

## Lemma

Suppose DES is an *ideal cipher* ( $2^{56}$  random invertible functions  $\pi_i : \{0, 1\}^{64} \rightarrow \{0, 1\}^{64}$ )

then  $\forall x, y$  there is at most *one* key  $k$  such that

$$y = DES(k, x)$$

with probability  $\geq 1 - 1/256 \approx 99.5\%$ .

## Exhaustive Search for block cipher key

- For two DES pairs :

$$(x_1, y_1 = \text{DES}(k, x_1)) \text{ and } (x_2, y_2 = \text{DES}(k, x_2))$$

unicity probability  $\approx 1 - 1/2^{71}$ .

- For AES-128: given two input/output pairs, unicity prob.  
 $\approx 1 - 1/2^{128}$
- Thus two input/output pairs are enough for Exhaustive Key Search.

## DES challenge

msg = "The unknown messages is : XXXX ... "  
CT =            y1            y2            y3            y4

**Goal:** find key  $k \in \{0, 1\}^{56}$  such that  $\text{DES}(k, x_i) = y_i$  for  $i = 1, 2, 3$ .

- 1997: DESCHALL project with internet search – 96 days
- 1998: EFF machine (DeepCrack) – 3 days        (250K \$)
- 1999: combined search – 22 hours
- 2006: COPACOBANA (120 FPGA) – 7 days        (10K \$).

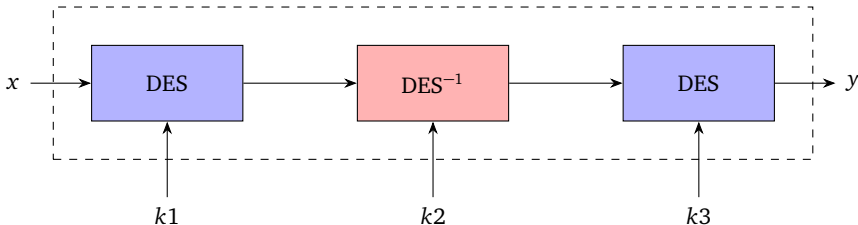
56 bit ciphers should not be used !!        128-bit key  $\Rightarrow 2^{72}$  days.

## 3DES Encryption and Key Selection

- 3DES uses **Encryption-Decryption-Encryption (EDE)** mode:

$$y = \text{DES}_{k_3} ( \text{DES}_{k_2}^{-1} ( \text{DES}_{k_1}(x) ) )$$

- Supports **single DES** when  $k_1 = k_2 = k_3$ .



## Security and Alternatives

- **3DES Security**: Resistant to brute-force but limited to  $2^{20}$  blocks per key.
- Efficient in hardware, slower in software.
- No 2DES due to the **meet-in-the-middle attack**.
- DESX: Strengthens DES with key whitening:

$$y = \text{DES}_k(x \oplus k_1) \oplus k_2$$

- Improves resistance to key search attacks.

# Introduction to AES

- **Algorithm of choice:** Widely used in numerous applications for secure encryption.
- **Key Lengths:** AES offers three key lengths—128, 192, and 256 bits.
- **Security:**
  - AES is resistant to brute-force attacks for several decades.
  - No known successful analytical attacks on AES.
  - **Quantum Threat:** AES-128 could be vulnerable to quantum attacks, while AES-192 and AES-256 are expected to resist them.



25  
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